

VU Medical Zone

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Immunology

BT 302 MID & FINAL TERM M.C.Q

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1. _____ components or barriers of immunity against infectious disease. **Three**
2. **Humoral** barrier mediated the inflammatory process.
3. **Lactoferrin & transferrin** ability to bind with **iron**, deprive bacteria from iron.
4. **Interferon** inhibits viral replication in cells.
5. **Lysozyme** has ability to lyses bacterial cell wall after digestion **Peptidoglycan**.
6. **Interleukin** has ability **induce fever & antimicrobial** acute phase protein.
7. In **inflammatory** process during infections, under the action various humoral substances immune cells recruit toward the site of infection.
8. **Neutrophils** phagocytes the invading agents & kill intracellularly.
9. **Macrophages** are involved in Phagocytosis & intracellular killing.
10. **Natural killer** has ability to role in **tumor surveillance** and has ability to kill **viral infected cells**.
11. **Eosinophils** also cause the cytotoxicity of parasitic infected cells.
12. The process of engulfing the invading or infectious agent by phagocyte. **Phagocytosis**
13. Pseudopod extends around the bacterium to form a vesicle called as **phagosome**.
14. Phagosome with trapped bacteria fuse with the secretory vesicles of phagocytes called **Lysosome**.
15. After the Phagocytosis the bacteria is ingested or killed by the process called **intracellular killing**.
16. **C1** protein is activated after interacting with **Fc** region of **IgG or IgM**.
17. In alternative pathway for complement activation C3 is directly converted in the absence of **Antibodies**.
18. **Mannose binding Lectins (MBL)** in serum bind to surface of the pathogen containing **mannose** in the cell wall.
19. Which one of the following pathology problems related to complementary system? **Capillary dilation and allergic reaction**
20. Primary Lymphoid organs are involved in maturation of **lymphocytes**.
21. **1 – 2 %** of lymphocytes are recirculating in every hour.
22. Which one of the following plays an important role in cell mediated immunity? **T- Lymphocytes**
23. A substance that reacts with the product of specific immune response is called **Antigen**.
24. Which one of the following induces specific immune response? **Immunogen**
25. A substance that is non-immunogenic but can react with product of specific immune response. **Hapten**
26. 1.Pain, heat, redness, swelling & loss of tissue functions are... **Signs of inflammation**

27. Diabetes Mellitus is... **non-infectious disease**
28. Migration of phagocytes across the capillary wall ... **Diapedesis**
29. For alternate pathway for complement activation **no need of Antibody**
30. Lymph nodes, spleen and Peyer's patches are **Secondary lymphoid organs**
31. By lymphocytes migration chances of successful antigen encounter may **increase**
32. **Hapten** is a substance that is non-immunogenic but can react with products of specific immune response
33. Very young and old individuals have **diminished immune response**
34. B-cells are activated directly without the help of T-cells for antibodies production in **T-independent antigens**
35. Fab and Fc are two fragments of **Antibodies**
36. Second most common serum antibody **IgA**
37. Humans can make **10^7 - 10^8** different Ab molecules
38. T-cells maturation in **Thymus**
39. Complement system have **complement proteins**
40. Complement proteins are activated by pathways **3**
41. Mostly IgA is present in secretions **saliva, tears, mucus & colostrums**
42. Particulate form of Ag **Agglutination reactions**
43. Particulate nature of antigen causes the reaction of antibody to form visible clumps **Agglutination**
44. Ag & Ab reaction which can be detected on the basis of substrate utilization by enzyme **ELISA**
45. Recombination involves rearrangement of DNA in somatic cells **Somatic recombination**
46. Variable and constant region are found in _____ chain of antibody. **Both heavy and light chain**
47. Hydrogen peroxide, Superoxide, hydroxyl radicals (OH), Activated halides and Nitrous oxide are the Antibacterial compounds released by the _____ **Macrophages**
48. Antigen presenting cell have major histocompatibility _____ molecules. **MCH class II**
49. Mature helper T cell express _____ ligand on their surface. **CD-40**
50. Antibody's heavy chain is encoded by the first constant region gene that is known as the _____ **the μ segments**
51. Helper T cells stimulate _____ to make antibodies. **B cells**
52. Macrophages _____ the microorganisms. **Engulf and digest**
53. Cellular immunity is mediated by _____ which target the cells infected by the pathogens. **T lymphocytes**
54. Interaction of antibodies and antigen provides the _____ response. **Specific**

55. C3 convertase is the complex of _____ **C4b-2b**
56. One of the following is released into the fluid surroundings where it works as a potent anaphylatoxin. **C5a**
57. _____ Cells phagocytose antibody-coated pathogens. **Neutrophils**
58. A viral protein made in an infected cell is degraded into fragments and _____ by a class I MCH protein. **Destroyed**
59. Antibody is composed of _____ polypeptide chain. **4**
60. Dendritic cells activate _____ immune system. **Adaptive**
61. Most of the components of the complement system are inactive until they are cleaved by a _____ **Protease**
62. TH-2 cells stimulate memory B cells to change the type of _____ that they secrete. **Antibody**
63. When activated by an **antigen presenting cells** naïve helper T cell can differentiate into either of two distinct types of effectors helper cell.
64. Dendritic cell secretes _____ to guide helper T cell to differentiate into Th 1(helper T cell type I). **IL-12**
65. Bone marrow performs which of the following functions? (Hematopoiesis, B cell maturation, B cell selection) **All**
66. The molecules of the microbes targeted by the immune system are _____ **pathogen associated molecular patterns.**
67. Complement system consists of around _____ protein. **30**
68. Co-receptor protein (CD28) on the surface of the T cell recognized _____ on the surface of the antigen presenting cell (APC).(CD80 and CD86, B7 protein, co stimulatory protein) **All**
69. Antibody are composed of four polypeptides chains, that is _____ **2 heavy and 2 light chain**
70. Mast cells Release **histamine** when damaged.
71. Lymph Nodes, contain a variety of _____ **WBCs**
72. Various heavy chains give distinctive conformation to _____ region of antibodies. (Hinge and tail) **Both**
73. The only antibody that can pass from mother to fetus _____ **IgG**
74. Several microorganism can be destroyed by enzyme found in (Saliva, tear, All, gastric juice) **All**
75. In 1970(WHO) define immunity as a immune response _____ **Antigen**
76. Damaged _____ cells releases histamine. **Mast cell**
77. _____ filters out older RBCs. **Spleen**
78. Membrane attack complex is formed by _____ **Cleavage of C5**
79. IgM or cold agglutins are active at _____ **4 degree centigrade**
80. Myasthenia Grave leads death due to _____ **respiratory failure**
91. After endocytosis of viral genome in infected cell next step _____

92. _____ develops in the bone marrow. **B cell**
93. _____ immunoglobulin is secreted as a monomer. **IgG**
94. Effector cytotoxic T cell directly kills cells infected with _____ (virus, bacteria, all, parasite) **Virus**
95. In MHC-I proteins, the side chains of amino acids of the peptide point outward, in a position to be recognized by receptors on _____ (helper T cell, APC, cytotoxic T cell, none) **cytotoxic T cell**
96. Helper T cell are activated by _____ (APC, effectors, activator, all) **APC**
97. Approximately 30% of all human tumors contain mutated version of _____ gene. **RAS**
98. Which one of the following is secondary lymphoid organ.
99. Immunoglobulin genes have relatively _____ **high mutation rate**
100. _____ is the defensive barrier that hinders pathogen invasion. **Skin**
101. C3b activates the _____ **C5**
102. Activated retinoblastoma (RB) protein prevents cells from advancing from _____ **G1 to S**
103. Role of NK cells in innate immunity NK- natural killer killing of cells infected by _____ pathogens (e.g. viruses) and tumor cells. **Intracellular**
104. Splicing of primary RNA transcript removes the transcript of any interons and extra _____ genes. **J segment.**
105. Rheumatoid arthritis is caused by the formation of autoantibody of type _____, called as rheumatoid factor. **IgM type**
106. The synovial fluid of patients contains increased no. of T-cells & macrophages in one of the following diseases. **Rheumatoid arthritis**
107. The idea of vaccines originated when a British physician Edward Jenner, who in 1796 used the _____ virus (Latin variola vaccinia) to confer protection against smallpox. **Cowpox**
108. For Anthrax _____ vaccine is used. **Inactivated**
109. One of the following is a T-cell associated auto immune disease. **Hashimoto's thyroiditis**
110. Killed/ inactivated vaccines contain killed, but previously _____ microorganism. **Virulent**
111. Vaccine are dead or _____ or purified products derived from them. **inactivated organisms**
112. For influenza _____ vaccine is used. **Live attenuated/ activated**
113. First vaccine in 1885 was developed by the French microbiologist Louis Pasteur roux the against _____ **Rabies**
114. The current vaccine for the streptococcus pneumonia, which causes pneumococcal pneumonia, consist of 23 antigenically different _____ **Capsular polysaccharides**
115. The first such recombinant antigen vaccine approved for human use is the _____ vaccine. **Hepatitis B**

116. One of the following purified macromolecule can be used as vaccines. **All of the given (recombinant microbial antigen, capsular polysaccharides, toxoids)**
117. In the case of the Myasthenia Gravis, nerve impulse cannot be transmitted from the nerves to the muscles because _____ cannot be produced. **Acetylcholine**
118. Appearance of blood red spots over the bridge of nose & cheeks is the characteristic features of ____ **Systemic Lupus Erythematosus (SLE)**
119. Myasthenia Gravis can be treated with _____ **Choline esterase inhibitors**
120. The characteristic features of system lupus Erythematosus is _____ **A mature neutrophil in the blood & bone marrow.**
121. Lupes Erythematosus can be treated with _____ (Corticosteroids, Antimitotic drugs, immune-suppressive drugs) **All of given**
122. Hashimoto's thyroiditis is characterized by the _____ of TH(thyroxin). **Deficiency**
123. Anti- inflammatory drugs can be used for the treatment of _____ **Rheumatoid Arthritis**
124. Some of the Attenuated vaccines contain live, attenuated microorganisms, or closely related but _____ dangerous organisms to produce a broad immune response. **less**
125. _____ cell develop into macrophages. **Monocytes**
126. Humoral immunity is mediated by soluble antibody proteins produced by _____ **B cells**
127. Can human make antibody molecules in the absence of **antigen**
128. T cell has capability to eliminate _____ **intracellular pathogens**
129. Type of blood cells that are responsible for eating foreign particles by engulfing them are _____ **WBCs**
130. Pluripotent hematopoietic stem cells found in _____ **bone marrow**
131. Immunological agents that modifies affect of other agents such as drug or vaccines _____ **adjuvant**
132. Hyper accurate rejection occur due to _____ **pre-existing antibodies**
133. After endocytosis of vital genome infected cell next step is _____ **intracellular killing**
134. When antibody binds to pathogens they can _____ pathogen. (**Active**, inactive, none, no effect).
135. C3 covertase is a complex of C3b and _____ **C4b**